Assignment

Chapter 1: Overviews

- **2.1** Suppose that we have a mutliprogrammed computer in which each job has identical characteristics. In one computation period, T, for a job, half the time is spent in I/O and the other half in processor activity. Each job runs for a total of N periods. Assume that a simple round-robin priority is used and that I/O operations can overlap with processor operation. Define the following quantities:
  - Turnaround time= actual time to complete a job.
  - Throughput= average number of jobs completed per time period T.
  - Processor utilization= percentage of time that the processor is active

Chapter 1

- Compute these quantities for one, two, and four simultaneous jobs, assuming that the period T is distributed in each of the following ways:
  - a.I/O first half, processor second half.
  - b.I/O first and fourth quarters, processor second and third quarters.

Assignment

Chapter 2: Process Description and Control

- **3.1** A system adopts a priority-based preemptive scheduling where the initial priority of a process increases by 1 after every 5 ms. In a recorded time span, the system has four process, P1, P2, P3 and P4, as shown in the follows table.

Process ID	Initial Priority	Arrival Time in MS	Total CPU Time in MS
P1	1	0	15
P2	3	5	7.5
P3	2	10	5
P4	2	15	10

Chapter 2

- Draw a timing diagram similar to Figure 3.7 and find the turnaround time for each process. Assume that the dispatcher takes 2.5 milliseconds for a process switch.
- **3.2(Seventh Edition)** Assume that at time 5 no system resources are being used except for the processor and memory. Now consider the following events as table. For each time 22, 37, and 47, identify which state each process is in. If a process is blocked, further identify the event on which is it blocked.

Chapter 2

Time	Events
At time 5:	P1 executes a command to read from disk unit 3.
At time 15:	P5's time slice expires.
At time 18:	P7 executes a command to write to disk unit 3.
At time 20:	P3 executes a command to read to disk unit 2.
At time 24:	P5 executes a command to write to disk unit 3.
At time 28:	P5 is swapped out.
At time 33:	An interrupt occurs from disk unit 2: P3's read is complete.
At time 36:	An interrupt occurs from disk unit 3: P1's read is complete.
At time 38:	P8 terminates.
At time 40:	An interrupt occurs from disk unit 3: P5's write is complete.
At time 44:	P5 is swapped back in.
At time 48:	An interrupt occurs from disk unit 3: P7's write is complete.